The Tilting Functor for Perfectoid Fields

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Abstract

This note was written as a final project for MAT 444 at Grinnell College, and comprises my first attempt at understanding perfectoid fields.

We define perfectoid fields and construct the tilting functor. We prove Scholze's generalization of the Fontaine-Wintenberger theorem: From the perspective of Galois theory, a perfectoid field K is indistinguishable from its tilt $K^\flat$. Our approach to untilting is that of Kedlaya and Liu [3], and relies on the machinery of Witt vectors, which we briefly sketch.

Notation & Terminology

- Fix, once and for all, a prime number p. In principle, most definitions we state could have a "(p-)" prepended to them (e.g., (p-)perfectoid); we leave this detail implicit.

- For a ring R of characteristic p, the *Frobenius* is the endomorphism $F : R \longrightarrow R,\ x \mapsto x^p$. Such a ring R is *perfect* if F is an automorphism.

- A ring A is *p-adically complete* if $A = \varprojlim A/(p^n)$. *Note: In this case and in others, we use the symbol "=" to mean that there exists a canonical isomorphism.*

- The *p-adic topology* on such a ring A is the one induced by the inverse limit, *i.e.* the coarsest topology such that the projection maps $A \longrightarrow A/(p^n)$ are continuous; equivalently, it is the one obtained by declaring the sets $p^n A$ to be a local base at 0.

- A *non-archimedean field* is a field K equipped with a non-trivial non-archimedean absolute value $|\cdot|$ with respect to which it is complete. We denote the ring of integers of a non-archimedean field K by $\mathfrak{o}_K$; this ring is local, and we denote its residue field by $\kappa_K = \mathfrak{o}_K/\mathfrak{m}_K$, where $\mathfrak{m}_K = \{x \in K : |x| < 1\}$ is the unique maximal ideal of $\mathfrak{o}_K$.

1. Tilting

Lemma 1.0.1. Let A be a ring. If $x, y \in A$ satisfy $x \equiv y \bmod p$, then $x^{p^n} \equiv y^{p^n} \bmod p^{n+1}$ for all $n \geqslant 0$.

Proof, following [1]. The case $n = 0$ is given. If we know that $x^{p^{n-1}} \equiv y^{p^{n-1}} \bmod p^n$ for some $n-1 \geqslant 0$, then we can write $x^{p^{n-1}} = y^{p^{n-1}} + p^n z$ for some $z \in A$. Therefore,

$$x^{p^n} = (y^{p^{n-1}} + p^n z)^p = \sum_{i=0}^{p} \binom{p}{i} (y^{p^{n-1}})^i (p^n z)^{p-i} = y^{p^n} + \sum_{i=1}^{p-1} \binom{p}{i} p^{n(p-i)} (y^{p^{n-i}})^i z^{p-i} + p^{pn} z^p.$$

Because p divides $\binom{p}{i}$ and $n(p-i) \geqslant n$ for all $1 \leqslant i \leqslant p-1$, the middle summand vanishes mod p^{n+1}. Likewise, because $pn \geqslant n+1$, the last summand does the same, so we have $x^{p^n} \equiv y^{p^n} \bmod p^{n+1}$. The result follows inductively. $\square$

Construction 1.0.1 (Perfection, following [1])**.** Let A be a p-adically complete ring. Let

$$R^{\mathrm{perf}} = \varprojlim_{F} A/(p) = \left\{ \overline{x} = (\overline{x}_n) \in \prod_{n=0}^{\infty} A/(p) : \overline{x}_{n+1}^p = F(\overline{x}_{n+1}) = \overline{x}_n \text{ for all } n \geqslant 0 \right\};$$

by construction, R^{perf} is a perfect ring of characteristic p; the inverse of Frobenius on R^{perf} is given by the shift map $(\overline{x}_n) \mapsto (\overline{x}_{n+1})$. From this point of view, it is not clear if R^{perf} has zerodivisors. On the other hand, we have a multiplicative monoid

$$A^\flat = \varprojlim_{x \mapsto x^p} A = \left\{ x \in \prod_{n=0}^{\infty} A : x_{n+1}^p = x_n \text{ for all } n \geqslant 0 \right\},$$

which does not have an immediate additive structure in general. However, component-wise reduction mod p induces a multiplicative map $A^\flat \longrightarrow R^{\mathrm{perf}}$. This is an isomorphism of multiplicative monoids: For injectivity, suppose we have $x, y \in A^\flat$ with $x_n \equiv y_n \bmod p$ for all n. By Lemma 1.0.1, we have $x_n = x_{n+k}^{p^k} \equiv y_{n+k}^{p^k} = y_n \bmod p^{k+1}$ for all $k \geqslant 0$. Because A is p-adically complete, it follows that $x_n = y_n$ for all n, hence $x = y$. As for surjectivity, let $\overline{x} \in R^{\mathrm{perf}}$, and fix a lift $x_n \in A$ of $\overline{x}_n \in A/(p)$ for each n. Then

$$x_{n+k+1}^p \equiv \overline{x}_{n+k+1}^p = \overline{x}_{n+k} \equiv x_{n+k} \mod p,$$

so by Lemma 1.0.1 again, $x_{n+k+1}^{p^{k+1}} \equiv x_{n+k}^{p^k} \bmod p^{k+1}$. Therefore, the sequence $(x_{n+k}^{p^k})_{k \geqslant 0}$ is Cauchy in A, hence converges to some $\widetilde{x}_n \in A$ for each n. Then $x_{n+k+1}^{p^{k+1}} - x_{n+k}^{p^k}$ converges to both 0 and $\widetilde{x}_{n+1}^p - \widetilde{x}_n$ as $k \to \infty$, so the $\widetilde{x}_n$'s define an element $\widetilde{x} \in A^\flat$. Finally, by construction, $\widetilde{x}_n \equiv \overline{x}_n \bmod p$ for all n, so that our natural map sends $\widetilde{x}$ to $\overline{x}$.

We can transfer the additive structure of R^{perf} to $A^\flat$ along this multiplicative bijection; we call the ring $A^\flat$ the *tilt* of A. To get a handle on addition in $A^\flat$, let $x, y \in A^\flat$. Then

$x+y$ is the unique element of $A^\flat$ reducing to $\overline{x}+\overline{y} = (\overline{x}_n+\overline{y}_n) \bmod p$. Since x_n+y_n lifts $\overline{x}_n+\overline{y}_n$ for all n, the argument of the previous paragraph shows that

$$(x+y)_n = \lim_{k\to\infty}(x_{n+k}+y_{n+k})^{p^k}.$$

It is also now clear that R^{perf} has no zerodivisors when A is a domain. After all, the multiplicative structure on R^{perf} is the same as that of $A^\flat$, and in the latter, if $xy=0$, then $0=(xy)_0 = x_0y_0 \in A$, so either $x_0=0$ or $y_0=0$ since A is a domain; without loss, suppose $x_0=0$. Then for all $n \geqslant 0$, $x_n^{p^n}=x_0=0$, so $x_n=0$, hence $x=0$.

Definition 1.0.1. A non-archimedean field K is *perfectoid* if $|K^\times|$ is not discrete, the residue field κ_K has characteristic p, and the residue ring $\mathfrak{o}_K/(p)$ is *semi-perfect*, *i.e.* has surjective Frobenius.

Let K be perfectoid. Because the residue field $\kappa_K=\mathfrak{o}_K/\mathfrak{m}_K$ has characteristic p, the maximal ideal $\mathfrak{m}_K\subseteq\mathfrak{o}_K$ contains p, meaning $|p|<1$. Given any family of morphisms $\{\varphi_n : A \longrightarrow \mathfrak{o}_K/(p^n)\}_{n\geqslant 1}$ commuting with the maps $\mathfrak{o}_K/(p^{n+1}) \longrightarrow \mathfrak{o}_K/(p^n)$, we know that $\varphi_{n+1}(a)\equiv\varphi_n(a) \bmod p^n$ for all n, so fixing lifts $x_n\in\mathfrak{o}_K$ for $\varphi_n(x)\in\mathfrak{o}_K/(p^n)$, the resulting sequence (x_n) satisfies $|x_{n+1}-x_n|<|p|^n$, hence is Cauchy in $\mathfrak{o}_K$. Therefore, $x_n\to\varphi(a)\in\mathfrak{o}_K$. From the continuity of the ring operations, it is easy to see that this yields a ring morphism $\varphi : A\longrightarrow\mathfrak{o}_K$, and that φ satisfies $\varphi(a)\equiv\varphi_n(a) \bmod p^n$ for all n. If we have any other such morphism $\psi : A\longrightarrow\mathfrak{o}_K$, then $|\varphi(a)-\psi(a)|\leqslant|p|^n$ for all n, and letting $n\to\infty$ shows $\varphi(a)=\psi(a)$ for all $a\in A$. This proves that $\mathfrak{o}_K$ is p-adically complete. Moreover, the topology on $\mathfrak{o}_K$ agrees with the p-adic topology: It certainly refines the p-adic one, and the balls of radius $|p|^n$ form a local base at 0 since $|p|^n\to 0$ as $n\to\infty$. Therefore, we can define the *tilt* of K to be the field $K^\flat=\mathrm{Frac}(\mathfrak{o}_K^\flat)$.

Example 1.0.1. A perfectoid field of characteristic p is precisely a perfect non-archimedean field with non-discrete value group. The key point is that for non-archimedean field K of positive characteristic, the ideal of $\mathfrak{o}_K$ generated by p is trivial, so that $\mathfrak{o}_K/(p)=\mathfrak{o}_K$. If the Frobenius is surjective on $\mathfrak{o}_K$, then since every element of K is of the form x/y for $x,y\in\mathfrak{o}_K$, in fact K has surjective Frobenius as well. But the Frobenius on K, hence on $\mathfrak{o}_K$, is automatically injective, so from this we see that semi-perfectness is equivalent to perfectness for $\mathfrak{o}_K$. Therefore, for a perfectoid field K of characteristic p,

$$(\mathfrak{o}_K/(p))^{\mathrm{perf}} = \left\{x\in\prod_{n=0}^{\infty}\mathfrak{o}_K : x_{n+1}^p = x_n \text{ for all } n\geqslant 0\right\} = \left\{(F^{-n}(x_0)) : x_0\in\mathfrak{o}_K\right\}\cong\mathfrak{o}_K,\text{[1]}$$

which shows that $K^\flat\cong K$. In other words, tilting is trivial in positive characteristic.

Example 1.0.2. Consider the completion $K=\widehat{\mathbb{Q}_p(p^{1/p^\infty})}$ of $\mathbb{Q}_p(p^{1/p^\infty})=\bigcup_{n\geqslant 1}\mathbb{Q}_p(p^{1/p^n})$. For each $n\geqslant 1$, the extension $\mathbb{Q}_p(p^{1/p^n})/\mathbb{Q}_p$ is totally ramified of degree n, so that the value group $|\mathbb{Q}_p(p^{1/p^n})^\times| = p^{\frac{1}{n}\mathbb{Z}}$. Thus, $|K^\times|=p^{\mathbb{Q}}$ is not discrete. The ring of integers of

[1] Here, the isomorphism sends $(F^{-n}(x_0))$ to $F^{-0}(x_0)=x_0$.

$\mathbb{Q}_p(p^{1/p^n})$ is just $\mathbb{Z}_p[p^{1/p^n}]$, so $\mathfrak{o}_K = \widehat{\mathbb{Z}_p[p^{1/p^\infty}]}$ is the completion of the ring $\mathbb{Z}_p[p^{1/p^\infty}] = \bigcup_{n\geqslant 1}\mathbb{Z}_p[p^{1/p^n}]$. Then $\mathfrak{o}_K/(p) = \mathbb{F}_p[t_1, t_2, \ldots]/(t_1^p, t_2^p - t_1, t_3^p - t_2, \ldots)$. Because $\mathbb{F}_p$ is perfect, and because each generator t_n is the image under Frobenius of t_{n+1}, the ring $\mathfrak{o}_K/(p)$ is semi-perfect. Therefore, K is perfectoid. One can show that the field $L = \widehat{\mathbb{Q}_p(\zeta_{p^\infty})}$ obtained by completing the extension of $\mathbb{Q}_p$ by all p-power roots of unity is perfectoid as well, and that $K^\flat = \widehat{\mathbb{F}_p((t))(t^{1/p^\infty})} = L^\flat$; for details, see [3, Example 1.3.6.].

At the moment, tilts of perfectoid fields have no analytic structure. To remedy this, we want to give $K^\flat$ an absolute value. To this end, define the *sharp map* $\sharp : K^\flat \longrightarrow K$ to be the multiplicative map given by the projection $x/y \mapsto x_0/y_0$, and let $|\cdot|_\flat : K^\flat \longrightarrow \mathbb{R}_{\geqslant 0}$ be the map with $|x|_\flat = |x^\sharp|$ for all $x \in K^\flat$.

Proposition 1.0.1. Let K be a perfectoid field. Then $|\cdot|_\flat$ is an absolute value on $K^\flat$ making the latter into a perfectoid field. In fact, $\mathfrak{o}_{K^\flat} = \mathfrak{o}_K^\flat$ and $|K^{\flat\times}|_\flat = |K^\times|$.

Proof, following [7]. As a function on $K^\flat$, it is clear that $|\cdot|_\flat$ is multiplicative, that it sends 1 to 1, and that the fiber over 0 is just 0 itself. What remains, then, is to show that it satisfies the ultrametric inequality. We first verify this for $x, y \in \mathfrak{o}_K^\flat$, where we understand addition. Here, we have

$$|x+y|_\flat = |(x+y)^\sharp| = \lim_{n\to\infty} |x_k + y_k|^{p^k} \leqslant \lim_{n\to\infty} \max\left\{\left|x_k^{p^k}\right|, \left|y_k^{p^k}\right|\right\} = \max\{|x|_\flat, |y|_\flat\}.$$

As for $x/w, y/z \in K^\flat$, we know that $|zx + wy|_\flat \leqslant \max\{|zx|_\flat, |wy|_\flat\}$; dividing by $|wz|_\flat$ yields the desired inequality. Certainly, $|x|_\flat \leqslant 1$ for all $x \in \mathfrak{o}_K^\flat$, since $x^\sharp \in \mathfrak{o}_K$. On the other hand, if $|x|_\flat \leqslant 1$, then $x_0 = x^\sharp \in \mathfrak{o}_K$, and then $|x_n| = |x_0|^{1/p^n} \leqslant 1$ for all n, so that $x \in \mathfrak{o}_K^\flat$. This proves that $\mathfrak{o}_{K^\flat} = \mathfrak{o}_K^\flat$.

We still need to show that $K^\flat$ is complete with respect to $|\cdot|_\flat$. But this is straightforward: Let $(x^{(m)})$ be a Cauchy sequence in $\mathfrak{o}_{K^\flat}$. Then the sequence $(x_0^{(m)})$ is Cauchy in $\mathfrak{o}_K$. We know that

$$\begin{aligned}\left|x_n^{(m+1)} - x_n^{(m)}\right|^{p^n} &\leqslant \max\left\{\left|x_0^{(m+1)} - x_0^{(m)}\right|, \left|\sum_{i=1}^{p^n-1}\binom{p^n}{i}\left(x_n^{(m+1)}\right)^i\left(x_n^{(m)}\right)^{p^n-i}\right|\right\}\\ &\leqslant \max\left\{\left|x_0^{(m+1)} - x_0^{(m)}\right|, |p|^n\right\} \to 0 \text{ as } m \to \infty,\end{aligned}$$

the second inequality following from the fact that p divides $\binom{p^n}{i}$ for $1 \leqslant i \leqslant p^n - 1$. Therefore, $(x_n^{(m)})$ is Cauchy in $\mathfrak{o}_K$. As a result, for each n, we can let $x_n = \lim_{m\to\infty} x_n^{(m)} \in \mathfrak{o}_K$, and then since the p-power map is continuous, $x_{n+1}^p = x_n$ for all $n \geqslant 0$, so we have an element $x = (x_n) \in \mathfrak{o}_{K^\flat}$ to which $(x^{(m)})$ converges. More generally, if $(x^{(m)})$ is a Cauchy sequence in K, then we can choose $\alpha \in K^{\flat\times}$ such that $\alpha x^{(m)} \in \mathfrak{o}_{K^\flat}$ for all m. This then converges to some $y \in \mathfrak{o}_{K^\flat}$, and then $x^{(m)} \to \alpha^{-1}y \in K^\flat$. This shows that $K^\flat$ is complete.

Because $\mathfrak{o}_{K^\flat}$ has characteristic p, the ideal generated by p therein is just the zero ideal, so $\mathfrak{o}_{K^\flat}/(p) = \mathfrak{o}_{K^\flat}$, and we already know $\mathfrak{o}_{K^\flat}$ to be perfect. To complete the proof, we need to show that $|K^{\flat\times}|$ is not discrete; it suffices to show that $|K^{\flat\times}|_\flat = |K^\times|$. From the definition of $|\cdot|_\flat$, it follows immediately that $|K^{\flat\times}|_\flat \subseteq |K^\times|$. For the other direction, it suffices to show that $|K^\times| \cap [|p|,1] \subseteq |K^{\flat\times}|_\flat \cap [|p|,1]$, and we do this in two steps. To start, let $|y| \in |K^\times| \cap (|p|,1]$, so $y \in \mathfrak{o}_K$. Since $\mathfrak{o}_K/(p)$ is semi-perfect, we can choose an element $\overline{x} \in (\mathfrak{o}_K/(p))^{\text{perf}}$ such that $\overline{x}_0 \equiv y \bmod p$. There corresponds a unique $x \in \mathfrak{o}_{K^\flat}$ with $x_n \equiv \overline{x}_n \bmod p$ for all n. In particular, $x_0 \equiv \overline{x}_0 \equiv y \bmod p$, so we can write $x_0 = y + pz$ for some $z \in \mathfrak{o}_K$. Then $|x|_\flat = |y|$ since $|y| > |p|$, which proves $|K^\times| \cap (|p|,1] \subseteq |K^{\flat\times}|_\flat \cap (|p|,1]$. What remains to be shown is that $|p| \in |K^{\flat\times}|_\flat$; this is trivial if $|p| = 0$, so assume $|p| \neq 0$. Using the non-discreteness of $|K^\times|$, choose $x \in K$ with $1 < |x| < |p|^{-1/p}$, so $|p| < |p| \cdot |x|^p < 1$. By semi-perfectness of $\mathfrak{o}_K/(p)$, there are $y, z \in \mathfrak{o}_K$ with $px^p = y^p + pz$, and then $|px^p| = |y|^p$, so $|y/x| = |p|^{1/p}$, and $|p|^{1/p} \in |K^\times| \cap (|p|,1)$, so that by the above, there is an element $t^{1/p} \in K^{\flat\times}$ with $|t^{1/p}|_\flat = |p|^{1/p}$, hence $|t|_\flat = |p|$. This completes the proof. $\square$

Corollary 1.0.1. The map $\mathfrak{o}_{K^\flat} \longrightarrow \mathfrak{o}_K/(p)$ sending x to the residue of $x^\sharp$ is a surjective ring morphism with kernel generated by any $t \in \mathfrak{o}_{K^\flat}$ with $|t|_\flat = |p|$. In particular, $\mathfrak{o}_{K^\flat}/(t) = \mathfrak{o}_K/(p)$, hence $\kappa_K = \kappa_{K^\flat}$.

Proof. For any $x, y \in \mathfrak{o}_{K^\flat}$, we know that $(x_k + y_k)^{p^k} \equiv x_k^{p^k} + y_k^{p^k} = x_0 + y_0 \mod p$, so

$$(x+y)^\sharp = (x+y)_0 = \lim_{k\to\infty} (x_k + y_k)^{p^k} \equiv x_0 + y_0 = x^\sharp + y^\sharp \mod p.$$

This proves the suggested map is a ring morphism. Because the Frobenius on $\mathfrak{o}_K/(p)$ is surjective, the morphism $\mathfrak{o}_{K^\flat} \longrightarrow \mathfrak{o}_K/(p)$ sending x to the residue of $x^\sharp$ mod p is surjective. Fixing $t \in \mathfrak{o}_{K^\flat}$ with $|t|_\flat = |p|$, the kernel consists of those elements $x \in \mathfrak{o}_{K^\flat}$ with $|x|_\flat = |x^\sharp| \leqslant |p| = |t|_\flat$; these are precisely the multiples of t. $\square$

Construction 1.0.2 (Tilting Functor)**.** Let **Perf** be the category whose objects are perfectoid fields and whose morphisms $\varphi : K \longrightarrow L$ are bounded ring morphisms, *i.e.* ring morphisms for which there exists some $B > 0$ such that $|\varphi(x)|_L \leqslant B \cdot |x|_K$ for all $x \in K$. For such a φ, if $x \in \mathfrak{o}_K$, then $|\varphi(x)|_L^n = |\varphi(x^n)|_L \leqslant B|x^n|_K \leqslant B$. Thus, $|\varphi(x)|^n$ is bounded for all n, so $\varphi(x) \in \mathfrak{o}_L$ and φ restricts to a map $\mathfrak{o}_K \longrightarrow \mathfrak{o}_L$. Because ring morphisms are $\mathbb{Z}$-linear, φ sends $p\mathfrak{o}_K$ into $p\mathfrak{o}_L$, so we get an induced map $\overline{\varphi} : \mathfrak{o}_K/(p) \longrightarrow \mathfrak{o}_L/(p)$; passing to inverse perfections yields a morphism $(\mathfrak{o}_K/(p))^{\text{perf}} \longrightarrow (\mathfrak{o}_L/(p))^{\text{perf}}$, $(\overline{x}_n) \mapsto (\overline{\varphi}(\overline{x}_n))$. Lifting this along our canonical isomorphism, we get $\varphi^\flat : \mathfrak{o}_{K^\flat} \longrightarrow \mathfrak{o}_{L^\flat}$, $(x_n) \mapsto (\varphi(x_n))$. Because φ is a morphism of fields, it is injective, so $\varphi^\flat$ is injective as well and extends to a ring morphism $\varphi^\flat : K^\flat \longrightarrow L^\flat$. For any $x/y \in K^\flat$, we have

$$|\varphi^\flat(x/y)|_{L^\flat} = |\varphi(x_0/y_0)|_L \leqslant B \cdot |x_0/y_0|_K = B \cdot |x/y|_{K^\flat},$$

so $\varphi^\flat$ is bounded, hence a morphisms of perfectoid fields. The assignment $\varphi \mapsto \varphi^\flat$ is manifestly functorial, so we have constructed the *tilting functor* from **Perf** to itself. In fact, because $K^\flat$ always has characteristic p, and because tilting is trivial in positive characteristic (Example 1.0.1), the essential image of tilting consists precisely of perfectoid fields of characteristic p.

2. Untilting

Lemma 2.0.1. Let A be p-adically complete. Fix a *system of representatives* $\Lambda \subseteq A$ for $A/(p)$, *i.e.* for every $\overline{\alpha} \in A/(p)$, there is a unique $\alpha \in \Lambda$ with $\alpha \equiv \overline{\alpha} \bmod p$. Then every element $x \in A$, there is a sequence of elements $\alpha^{(n)} \in \Lambda$ such that $x = \sum_{n\geqslant 0} \alpha^{(n)} p^n$. If A is p-torsion-free, *i.e.* if p is not a zerodivisor on A, then the α_n's are uniquely determined by x.

Proof. Let $x \in A$, and let $\overline{\alpha^{(0)}}$ be the image of x in $A/(p)$. Then taking $\alpha^{(0)} \in \Lambda$ to be the corresponding representative, we have $x = \alpha^{(0)} + px'$ for some $x' \in A$. Repeating the argument with x', we get $\alpha^{(1)} \in \Lambda$ such that $x = \alpha^{(0)} + p\alpha^{(1)} + p^2 x''$. Continuing in this manner, we get a series representation of the desired form.

Now suppose A is p-torsion-free, and we have another sequence $\beta^{(n)} \in \Lambda$ such that $\sum_{n\geqslant 0} \alpha^{(n)} p^n = \sum_{n\geqslant 0} \beta^{(n)} p^n$. Reducing, we see that $\alpha^{(0)} \equiv \beta^{(0)} \bmod p$, so that $\alpha^{(0)} = \beta^{(0)}$ by uniqueness of lifts in Λ. Subtracting $\alpha^{(0)}$, we see that $\sum_{n\geqslant 1} \alpha^{(n)} p^n = \sum_{n\geqslant 1} \beta^{(n)} p^n$. Since A is p-torsion-free, we can divide by p to get $\sum_{n\geqslant 0} \alpha^{(n+1)} p^n = \sum_{n\geqslant 0} \beta^{(n+1)} p^n$. Repeating the argument shows $\alpha^{(n)} = \beta^{(n)}$ for all $n \geqslant 0$, completing the proof. □

Example 2.0.1 (Sharp Representatives)**.** By Corollary 1.0.1, the tilt $\mathfrak{o}_{K^\flat}$ surjects onto $\mathfrak{o}_K/(p)$ via a map factoring through $\sharp : \mathfrak{o}_{K^\flat} \longrightarrow \mathfrak{o}_K$. Thus, every element of $\mathfrak{o}_K/(p)$ has a lift in $\mathfrak{o}_K$ lying in the image of $\sharp$. Discarding redundant elements as needed, we see that the image of $\sharp$ contains a system of representatives as in Lemma 2.0.1, so every element of $\mathfrak{o}_K$ can be written as a series $\sum_{n\geqslant 0} (x^{(n)})^\sharp p^n$ for some $x^{(n)}$'s in $\mathfrak{o}_{K^\flat}$.

Definition 2.0.1. A p-adically complete ring A is a *strict p-ring* if it is p-torsion-free, meaning p is not a zerodivisor on A, and $A/(p)$ is perfect. We give a strict p-ring the p-adic topology by declaring the family $\{p^n A\}_{n\geqslant 1}$ to be a local base at 0; likewise, we require morphisms of strict p-rings to be continuous.

Proposition 2.0.1 ([3, Lemma 1.1.6. & Theorem 1.1.8.])**.** Let A be a strict p-ring and let B be p-adically complete. For every multiplicative map $t : A/(p) \longrightarrow B$ lifting a ring morphism $\bar{t} : A/(p) \longrightarrow B/(p)$, there is a unique p-adically continuous ring morphism $T : A \longrightarrow B$ making the following commute:

$$\begin{tikzcd} A \arrow[r, "T"] \arrow[d] & B \arrow[d] \\ A/(p) \arrow[r, "\bar{t}"] \arrow[ur, "t"] & B/(p) \end{tikzcd} .$$

Moreover, the functor from strict p-rings to perfect rings of characteristic p sending A

to $A/(p)$ is an equivalence of categories. In particular, for each perfect ring R, there is a unique (up to unique isomorphism) strict p-ring $W(R)$ with $W(R)/(p) = R$, called the *ring of Witt vectors with coefficients in R*.

Lemma 2.0.2. Let R be a perfect ring of characteristic p and let A be p-adically complete. Suppose we have a ring morphism $\bar{t} : R \longrightarrow A/(p)$. Then there is a unique multiplicative map $t : R \longrightarrow A$ lifting $\bar{t}$. In particular, $t(\bar{x}) \equiv x_n^{p^n} \bmod p^{n+1}$ for any $x_n \in A$ lifting $\bar{t}(\bar{x}^{1/p^n})$.

Proof. Denote the projection $A \longrightarrow A/(p)$ by π, and let $\bar{t} : R \longrightarrow A/(p)$ be a morphism of rings. By Lemma 1.0.1, there is a well-defined multiplicative map $\theta_n : A/(p) \longrightarrow A/(p^{n+1})$, $\bar{x} \mapsto \bar{x}^{p^n}$ for each n. Because R is perfect, the Frobenius F on R is invertible, so we have a family of multiplicative maps $\{\theta_n \circ \bar{t} \circ F^{-n} : R \longrightarrow A/(p^{n+1})\}_{n \geqslant 0}$. One quickly checks that these are coherent, so since $A = \varprojlim A/(p^{n+1})$, we get a unique multiplicative map $t : R \longrightarrow A$ agreeing with $\theta_n \circ \bar{t} \circ F^{-n} \bmod p^{n+1}$ for all n. In particular, $\pi \circ t = \theta_0 \circ \bar{t} \circ F^{-0} = \bar{t}$. For the second claim, let $\bar{x} \in R$, $n \geqslant 0$, and fix a lift $x_n \in A$ of $\bar{t}(x^{1/p^n})$. By our construction of t, we get $t(\bar{x}) \equiv (\theta_n \circ \bar{t} \circ F^{-n})(\bar{x}) = (\bar{t}(x^{1/p^n}))^{p^n} \equiv x_n^{p^n} \bmod p^{n+1}$, and we win. $\square$

Construction 2.0.1 (Canonical Coordinates)**.** We can describe the elements of $W(R)$ explicitly. Let R be a perfect ring of characteristic p. Taking $A = W(R)$ and $\bar{t} = \mathrm{id}_R$ in Lemma 2.0.2, we see that $\pi : W(R) \longrightarrow R$ has a unique multiplicative section $[\cdot] : R \longrightarrow W(R)$, the *system of canonical representatives*. We can describe $[\cdot]$ analytically as follows: Let $\bar{x} \in R$, and for each n, let $x_n \in W(R)$ be a lift of $\bar{x}^{1/p^n}$. Then, by the second part of Lemma 2.0.2, $[\bar{x}] \equiv x_n^{p^n} \bmod p^{n+1}$ for all n. As a result, $[\bar{x}] = \lim_{n\to\infty} x_n^{p^n}$ in the p-adic topology. Applying Lemma 2.0.1 to $\Lambda = [A/(p)]$, we see that every element of $W(R)$ can be written uniquely in the form $\sum_{n \geqslant 0} [\bar{\alpha}^{(n)}] p^n$ for some $\bar{\alpha}^{(n)}$'s in R. Moreover, every sequence of $\bar{\alpha}^{(n)}$'s yields an element of $W(R)$ in the way, precisely by the p-adic completeness of $W(R)$.

Let K be a perfectoid field. By Corollary 1.0.1, the sharp map $\sharp : \mathfrak{o}_{K^\flat} \longrightarrow \mathfrak{o}_K$ lifts a ring morphism $\mathfrak{o}_{K^\flat} \longrightarrow \mathfrak{o}_K/(p)$. Thus, by Proposition 2.0.1, there is a unique continuous morphism $\Theta_K : W(\mathfrak{o}_{K^\flat}) \longrightarrow \mathfrak{o}_K$ lifting $\sharp$.

Proposition 2.0.2. The morphism $\Theta = \Theta_K : W(\mathfrak{o}_{K^\flat}) \longrightarrow \mathfrak{o}_K$ is surjective.

Proof. To start, let $\bar{x} \in \mathfrak{o}_{K^\flat}$. Fixing a lift $x_n \in W(\mathfrak{o}_{K^\flat})$ of $\bar{x}^{1/p^n}$ for each n, we have

$$\Theta([\bar{x}]) = \Theta\left(\lim_{n\to\infty} x_n^{p^n}\right) = \lim_{n\to\infty} \Theta(x_n)^{p^n}$$

by the continuity of Θ. By construction, $\Theta(x_n) \equiv (\bar{x}^\sharp)^{1/p^n} \bmod p$. By Lemma 1.0.1, $\Theta(x_n)^{p^n} \equiv \bar{x}^\sharp \bmod p^{n+1}$ for all n, which proves $\Theta([\bar{x}]) = \bar{x}^\sharp$. Then for any $\sum_{n \geqslant 0} [\alpha^{(n)}] p^n \in W(\mathfrak{o}_{K^\flat})$, by the continuity and $\mathbb{Z}$-linearity of Θ,

$$\Theta\left(\sum_{n=0}^{\infty} [\alpha^{(n)}] p^n\right) = \sum_{n=0}^{\infty} \Theta([\alpha^{(n)}]) p^n = \sum_{n=0}^{\infty} (\alpha^{(n)})^\sharp p^n.$$

By Example 2.0.1, every element of $\mathfrak{o}_K$ can be written as a series of the form on the right, which proves the claim. $\square$

This suggests that the data lost in tilting can be recovered from Θ. The approach of Kedlaya and Liu to untilting consists of understanding which ideals arise as kernels of theta maps, and then to use such ideals to construct perfectoid fields of mixed characteristic from those of positive characteristic. It turns out that these ideals are of a very particular sort. For a perfectoid field F of positive characteristic, an element $z = \sum_{n\geqslant 0}[z^{(n)}]p^n \in W(\mathfrak{o}_F)$ is *primitive* if $|z^{(0)}|_F = p^{-1}$ and $(z - [z^{(0)}])/p \in W(\mathfrak{o}_F)^\times$. An ideal I of $W(\mathfrak{o}_F)$ is *primitive* if it is generated by a single primitive element. Write $\mathbf{Perf}^+$ for the category of pairs (F, I) of a perfectoid field F of characteristic p and a primitive ideal $I \subseteq W(\mathfrak{o}_F)$, with arrows $(E, I) \longrightarrow (F, J)$ consisting of morphisms of perfectoid fields $E \longrightarrow F$ whose induced map $W(\mathfrak{o}_E) \longrightarrow W(\mathfrak{o}_F)$ sends I into J. Because tilting is trivial in positive characteristic, we want to isolate the mixed characteristic case: Write $\mathbf{Perf}^{\text{mix}}$ for the category of characteristic 0 perfectoid fields and bounded morphisms.

Theorem 2.0.1 (Kedlaya-Liu, [3, Theorem 1.5.1.])**.** The functor from $\mathbf{Perf}^{\text{mix}}$ to $\mathbf{Perf}^+$ sending a perfectoid field K of characteristic 0 to $(K^\flat, \ker \Theta_K)$ and a morphism $\pi : K \longrightarrow L$ to the arrow corresponding to $\pi^\flat$ is an equivalence of categories with weak inverse sending (F, I) to $(W(\mathfrak{o}_F)/I)[p^{-1}]$[2].

Lemma 2.0.3. Let K be a perfectoid field of characteristic 0, and let $I = \ker \Theta_K$. For $F/K^\flat$ a finite extension, let $F^\sharp = (W(\mathfrak{o}_F)/I)[p^{-1}]$. Then $[F^\sharp : K] = [F : K^\flat]$. Moreover, if $F/K^\flat$ is Galois, then so is $F^\sharp/K$, and $\mathrm{Gal}(F^\sharp/K) = \mathrm{Gal}(F/K^\flat)$.

Proof, following [3]. Write $\ker \Theta_K = (z)$ for some primitive element $z \in W(\mathfrak{o}_{K^\flat})$. The field $K^\flat$ is perfectoid of characteristic p, hence perfect (Example 1.0.1), so the extension $F/K^\flat$ is separable. Let $\widetilde{F}$ be the Galois closure of F. Then $\widetilde{F}/F$ is a finite extension, so $\widetilde{F}$ has a natural absolute value extending that of F with respect to which it is complete; in particular $\widetilde{F}$ is a non-discretely-valued non-archimedean field. Moreover, because F is perfect, so is $\widetilde{F}$, whence it follows that $\widetilde{F}$ is perfectoid. We have an inclusion $\mathfrak{o}_F \hookrightarrow \mathfrak{o}_{\widetilde{F}}$, which induces an inclusion of Witt vector rings $W(\mathfrak{o}_F) \hookrightarrow W(\mathfrak{o}_{\widetilde{F}})$. Then $\widetilde{I} = zW(\mathfrak{o}_{\widetilde{F}})$ is a primitive ideal, and the morphism $(F, I) \longrightarrow (\widetilde{F}, \widetilde{I})$ yields a perfectoid extension $\widetilde{F}^\sharp = (W(\mathfrak{o}_{\widetilde{F}})/\widetilde{I})[p^{-1}]$ of K by Theorem 2.0.1.

Let H be a subgroup of $G = \mathrm{Gal}(\widetilde{F}/K^\flat)$. The action of H restricts to an action on $\mathfrak{o}_{\widetilde{F}}$, which induces an action on $W(\mathfrak{o}_{\widetilde{F}})$ fixing the subring $W(\mathfrak{o}_{K^\flat})$. Becauze $z \in W(\mathfrak{o}_{K^\flat})$, this action stabilizes the ideal $\widetilde{I}$, so the weak inverse to tilting from Theorem 2.0.1 yields an action of H on $\widetilde{F}^\sharp$ fixing K; equivalently, we have a group morphism $H \longrightarrow \mathrm{Gal}(\widetilde{F}^\sharp/K)$. Because said weak inverse is faithful, *i.e.* injective on hom-sets, this map of groups is injective, so we may view H as a subgroup of $\mathrm{Gal}(\widetilde{F}^\sharp/K)$. Concretely, for an element $x \in \widetilde{F}^\sharp$ lifted by $p^{-k}\sum_{n\geqslant 0}[\alpha^{(n)}]p^n \in W(\mathfrak{o}_{\widetilde{F}})[p^{-1}]$ and $\sigma \in H$, the same argument as in the proof of Proposition 2.0.2 shows

$$\sigma(x) = p^{-k}\sum_{n=0}^{\infty}[\sigma(\alpha^{(n)})]p^n.$$

[2]In particular, the ring $(W(\mathfrak{o}_F)/I)[p^{-1}]$ has a natural absolute value making it into a perfectoid field of characteristic 0; see [3, Theorem 1.4.13.]

Then by the uniqueness of canonical coefficients (Lemma 2.0.1), we see that x is fixed by H if and only if it has a lift $p^{-k}\sum_{n\geqslant 0}[\alpha^{(n)}]p^n$ with each $\alpha^{(n)} \in \widetilde{F}^H$, *i.e.* x lies in the quotient $(W(\mathfrak{o}_{\widetilde{F}^H})/\widetilde{I}\cap W(\mathfrak{o}_{\widetilde{F}^H}))[p^{-1}]$, which is naturally a subring of $\widetilde{F}^\sharp$ via the inclusion induced by $\widetilde{F}^H \hookrightarrow \widetilde{F}$. Observe that $\widetilde{I}\cap W(\mathfrak{o}_{\widetilde{F}^H}) = zW(\mathfrak{o}_{\widetilde{F}^H})$. Indeed, because H fixes $K^\flat$, $K^\flat \subseteq \widetilde{F}^H$, hence $z \in W(\mathfrak{o}_{K^\flat}) \subseteq W(\mathfrak{o}_{\widetilde{F}^H})$, which implies $zW(\mathfrak{o}_{\widetilde{F}^H}) \subseteq \widetilde{I}\cap W(\mathfrak{o}_{\widetilde{F}^H})$; the reverse inclusion follows from the uniqueness of canonical coordinates. Therefore, the fixed field $\widetilde{F}^{\sharp H} = (W(\mathfrak{o}_{\widetilde{F}^H})/zW(\mathfrak{o}_{\widetilde{F}^H}))[p^{-1}]$.

When $H = G$, we have $\widetilde{F}^G = K^\flat$. Then as the ideal $I = \ker\Theta_K$, the ring $(W(\mathfrak{o}_{K^\flat})/I)[p^{-1}] = K$ by Theorem 2.0.1, which proves that $\widetilde{F}^{\sharp G} = K$. Likewise, taking $\widetilde{G} = \mathrm{Gal}(\widetilde{F}/F)$, we have $\widetilde{F}^{\widetilde{G}} = F$, so that by a similar argument $\widetilde{F}^{\sharp\widetilde{G}} = F^\sharp$. By Artin's lemma [4, Corollary 3.11], the extension $\widetilde{F}^\sharp/\widetilde{F}^{\sharp H}$ is Galois of degree $|H|$ for all subgroups H of G. Therefore, $[\widetilde{F}^\sharp : K] = |G| = [\widetilde{F} : K^\flat]$ and $[\widetilde{F}^\sharp : L] = |\widetilde{G}| = [\widetilde{F} : F]$, so by the tower theorem,

$$[F^\sharp : K] = \frac{[\widetilde{F}^\sharp : K]}{[\widetilde{F}^\sharp : F^\sharp]} = \frac{[\widetilde{F} : K^\flat]}{[\widetilde{F} : F]} = [F : K^\flat],$$

which proves the claim about degrees. Finally, in the case where $F/K^\flat$ is Galois, so $F = \widetilde{F}$, we have shown that $F^{\sharp G} = K$, so $F^\sharp/K$ is Galois with group G by [4, Theorem 3.10]. □

Lemma 2.0.4 ([3, Lemma 1.5.4.])**.** *If K is a perfectoid field with algebraically closed tilt $K^\flat$, then K is algebraically closed.*

Theorem 2.0.2 (Scholze)**.** Let K be a perfectoid field. Then every finite extension L/K is perfectoid, and tilting $L \mapsto L^\flat$ induces an equivalence between the category of finite extensions of K and those of $K^\flat$.

Proof, following [3]. Because tilting in positive characteristic is trivial (Example 1.0.1), we need only consider the case of K a perfectoid field of characteristic 0.

The key remaining step is showing that finite extensions of K are perfectoid. Write $\ker\Theta_K = (z)$ for some primitive $z \in W(\mathfrak{o}_{K^\flat})$. Fix an algebraic closure $\overline{K^\flat}$ of $K^\flat$, and let $E^\flat$ be the corresponding completion. By [2, Theorem 1.1.], $E^\flat$ is algebraically closed as well. From the inclusion $W(\mathfrak{o}_{K^\flat}) \hookrightarrow W(\mathfrak{o}_{E^\flat})$, we see that z is a primitive element of $W(\mathfrak{o}_{E^\flat})$ as well, so we can apply Theorem 2.0.1 to the morphism $(K^\flat,(z)) \longrightarrow (E^\flat,(z))$ to get an extension E/K of perfectoid fields; by Lemma 2.0.4, E is algebraically closed. By Lemma 2.0.3, to each finite Galois extension $F/K^\flat$, there corresponds a finite Galois extension $F^\sharp/K$ of perfectoid fields with $F^{\sharp\flat} = F$ and $\mathrm{Gal}(F^\sharp/K) = \mathrm{Gal}(F/K^\flat)$. Let $\overline{F}^\sharp \subseteq E$ be the union of the subfields $F^\sharp$ corresponding to finite Galois extensions $F/K^\flat$. Then $\overline{F}^\sharp$ is perfectoid, and the tilt $\overline{F} = \overline{F}^{\sharp\flat}$ is the union of the finite Galois extensions of $K^\flat$. Because $K^\flat$ is perfect, every finite extension is separable, hence is contained in a finite Galois extension. It follows that $\overline{F} = \overline{K^\flat}$, which is dense in the completion $E^\flat$.

Let Z be the closure of $\overline{F}^\sharp$ in E; note that Z is a subring, because by the continuity of the ring operations in $E^\flat$, if x and y are limits of nets in $\overline{F}^\sharp$, then so are $x+y$ and xy. Then because the sharp map $\sharp : E^\flat \longrightarrow E$ is continuous, the preimage $Z' \subseteq E^\flat$ of Z under $\sharp$ is closed. Moreover, because $x^\sharp \in \overline{F}^\sharp$ for all $x \in \overline{F}$, Z' contains $\overline{F}$, hence $Z' = E^\flat$ by the density of $\overline{F}$. It follows that Z contains the image of $\sharp$, hence contains any term of the form $\sum_{n=0}^{N}(x^{(n)})^\sharp p^n$ because Z is a ring. But by Example 2.0.1, such terms are dense in E, hence $\overline{F}^\sharp$ is dense in E. If $f \in \overline{F}^\sharp[X]$ is a separable polynomial splitting as $\prod_i (X - \alpha_i)$ in the algebraically closed field E, then by density we can choose $\beta \in \overline{F}^\sharp$ with $|\beta - \alpha_1| < |\alpha_i - \alpha_1|$ for all $i \neq 1$, so by Krasner's lemma [5, p. 8.1.6.], $\alpha_1 \in \overline{F}^\sharp$; this shows that $\overline{F}^\sharp$ is separably closed. Because K has characteristic 0, every finite extension is separable. Thus, if L/K is a finite extension, then L is contained in $\overline{F}^\sharp$, hence in a finite perfectoid Galois extension $\widetilde{L}$ of K. By [4, Corollary 3.13], the extension $\widetilde{L}/L$ is Galois; let $G = \mathrm{Gal}(\widetilde{L}/L)$. Then from the proof of Lemma 2.0.3, we see that $L = \widetilde{L}^G$ is an untilt of $\widetilde{L}^{\flat G}$, hence is perfectoid.

At last, the equivalence from Theorem 2.0.1 restricts to an equivalence between finite extensions of K and the full subcategory of $\mathbf{Perf}^+$ whose objects are of the form $(F, (z))$ for $F/K^\flat$ a finite extension. The latter subcategory is clearly equivalent to the category of finite extensions of $K^\flat$ (by forgetting (z)), whence the correspondence follows. □

Note that if L/K is finite and Galois, then because tilting is an equivalence, $G = \mathrm{Gal}(L/K) = \mathrm{Aut}(L^\flat/K^\flat)$, and computing element-wise shows $(L^\flat)^G = (L^G)^\flat = K^\flat$, so $L^\flat/K^\flat$ is Galois.

Corollary 2.0.1 (Scholze, Fontaine-Wintenberger)**.** Let K be a perfectoid field. The absolute Galois groups $G_K = \mathrm{Gal}(\overline{K}/K)$ and $G_{K^\flat} = \mathrm{Gal}(\overline{K^\flat}/K^\flat)$ are naturally isomorphic as profinite groups.

Proof. Under tilting correspondence, Galois extensions of K correspond to Galois extensions of $K^\flat$, and by Lemma 2.0.3, $\mathrm{Gal}(L/K) = \mathrm{Gal}(L^\flat/K^\flat)$ for all L/K. Thus, the diagrams of finite quotients of G_K and $G_{K^\flat}$ are naturally isomorphic; [4, Example 7.26] yields

$$\mathrm{Gal}(\overline{K}/K) = \varprojlim_{\substack{L/K \\ \text{finite Galois}}} \mathrm{Gal}(L/K) = \varprojlim_{\substack{F/K^\flat \\ \text{finite Galois}}} \mathrm{Gal}(F/K^\flat) = \mathrm{Gal}(\overline{K^\flat}/K^\flat).$$

□

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